



POLITECNICO
MILANO 1863

DEPARTMENT
OF AEROSPACE SCIENCE
AND TECHNOLOGY

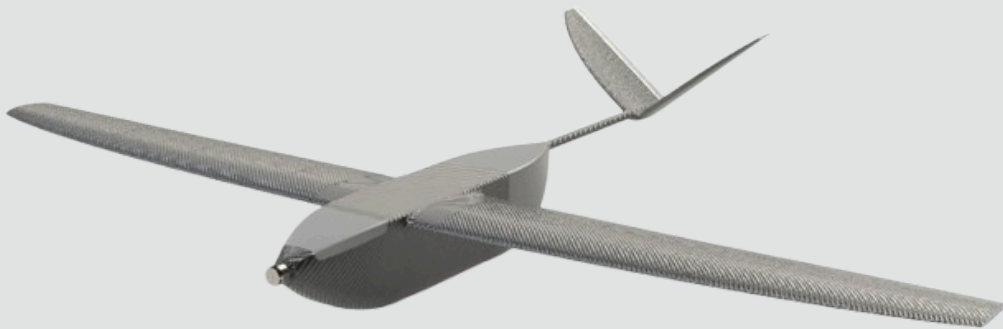
FLY MI

DECEMBER 2024

CRITICAL DESIGN REVIEW

13 - Fly-Mi - Politecnico di Milano

O.L.I.V.I.A. (Optimized Lightweight
Intelligent Vehicle for Immediate
Assistance)



TEAM MEMBERS

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ACADEMIC LEAD

Lorenzo Trainelli

CARACOL



Critical Design Review

Fly-Mi

Header

Team: Fly-Mi

Team Number: 13

University: Politecnico di Milano

Type of Meeting: Critical Design Review

Location: Online, Microsoft Teams

Participants

Present:

- Andrea Romano, Chartered Engineer
- Marta Catalano, Team Leader
- Filippo Coacci, CTO
- Paola Mansueti, Project Manager
- Luigi Hayashida, Vicepresident
- Federico Sottocorno, Treasurer
- Giorgia Pallara, Head of Structures Department
- Lorenzo Martella, Head of Structures Department
- Giuseppe Rizi, Head of Aerodynamics Department
- Lorenzo Lucatello, Head of Flight Mechanics Department
- Edoardo Semeria, Head of Flight Mechanics Department
- Alex Turcu, Head of Electronics Propulsion and Control Department

Agenda

1. Design Presentation
2. Requirements Review and Compliance Assessment
3. Reviewer's Feedback and Recommendations

Introduction

This meeting is the Critical Design Review (CDR), scheduled according to the GANTT chart. Although the timeline diagram provided in Section 2 of the regulation, titled *"Competition Overview,"* places the CDR after the Design and Development Specification, the guidance in Annex D, Section 4, titled *"Critical Design Review (CDR),"* states: *"This review is held at the end of your design phase and before you start major component manufacture and system testing."*

As the team has completed the design phase and the submission date for the CAD models of the moulds for lamination is approaching—marking the beginning of the manufacturing phase—it was decided to conduct the CDR to ensure compliance with the GANTT chart and the guidelines in the annex.

The review is being conducted by an engineer from a sponsoring company who has been involved in the project from the outset, assisting with project management and demonstrating a strong understanding of the competition's regulations. This individual is distinct from the Academic Lead who conducted the Preliminary Design Review (PDR) and meets the criteria specified in the aforementioned annex, which allows the CDR to be conducted by a "*Chartered Engineer*."

Discussion

- **Topic 1: Design Presentation**

- Summary:

The team presented the aircraft design through a structured presentation organized by departments. In sequence, the work carried out by the Aerodynamics, Flight Mechanics, Structures, Electronics Propulsion And Control departments was presented and analyzed. Each segment highlighted the contributions that led to the finalization of the design, with the respective department heads delivering the presentations.

- Key points:

- * Aerodynamics:

- Process Overview: The process for defining the airfoil and 3D wing was analyzed. The airfoil design involved three phases: "Database Skimming" using MATLAB, "Airfoil Optimization" with Xoptfoil2, and "Airfoil Selection" through XFLR5. The 3D wing design included "3D Drafts," "Geometry Definition," and "Wing Selection," using MATLAB and XFLR5.
 - Data: Tables and data were presented showing the main optimization improvements, including polar and airfoil profiles before and after optimization. Key wing parameters such as planform, surface area, span, aspect ratio, twist, taper ratio, and chord were detailed. The elliptical lift distribution graph and XFLR5 data validation through CFD simulations in Ansys were also shown.

- * Flight Mechanics:

- Stability and control analysis of the wing-tail combination by evaluating multiple geometries proposed by the Aerodynamics department, ensuring good stall characteristics, sizing flaps and ailerons iteratively and maximizing flap deflection to meet takeoff and landing performance requirements. Results validation and comparison with previous projects were shown.
 - Tail design by implementing an algorithm to minimize total mass and wetted area and creating a tail geometry database. The lightest tube-tail combination that satisfies predefined stability criteria was selected.
 - Aircraft balancing and tail optimization: initial estimation of mass and CG position revealed challenges in balancing the aircraft at the desired point. The issue was addressed by reducing the tube length while increasing the tail surface to maintain satisfactory stability parameters. Tail control surfaces were sized to ensure high control authority.
 - Data: Tables and charts showing control surfaces' dimensioning, static and dynamic stability parameters and tail geometry parameters were presented.

- * Structures:

- Dimensioning and Weight estimation: Tables showing the aircraft's parts dimensions and a detailed weight estimation were shown. The main characteristics of the chosen materials (Carbon fibre, Glass Fiber, Kevlar, Rohacell, Nomex, PCPBTGF filament) were presented.
 - Wing structure: definition of a 5-section structure, with 14 total ribs and 1 spar per wing. The wing-fuselage connection is made up of a carbon tube, a pin-blocking rotation and a drawbolt-blocking translation. Wing to wing connection is realised by a carbon tube and then the two parts will be glued together. The planned ply book for the wing and corresponding validation by finite element simulation were shown.

- Fuselage structure and ply book: a CAD rendering of the fuselage containing the formers and the wing connection structure was shown along with the ply book and moulds planning.
- Tail structure and ply book: presentation of a V tail structure with 3 ribs and a hub for the tail-tube connection. Tail-planned ply book and corresponding validation by finite element simulation are also shown.
- Landing Gear: the studies for the main landing gear geometry and C-section were presented along with the planned ply book and corresponding validation by finite element analysis. Front-wheel design is currently still in the process of definition.
- Payload Release system: a servoactuated pin will release a retention strap connected to a torsion spring, releasing the payload on the hatch doors, which will open due to the weight of the payload, and then close again thanks to a spring mechanism.
- * Electronics Propulsion And Control:
 - Electronic and propulsive components: A detailed table reporting the type, function, selection criteria, model and weight of each electronic and propulsive component was presented.
 - Autonomous flight test: the GPS image depicting the trajectories flown during the first autonomous flight test with manual landing, conducted on a different aircraft model, was presented.
 - Propulsion system sizing script: a MATLAB script to model the propulsion system was developed, enabling the preliminary selection of motors, propellers, and batteries for current and future drone projects. Using an object-oriented approach, components are modelled in a database to simulate system behaviour and meet mission requirements, enhancing code readability and scalability. Propeller and motor selection relies on coupling power curves derived from official tables, technical specifications, and experimental data. Future experimental validation will further improve the script's accuracy and reliability.
- Feedback:

It was noted that some design elements that were supposed to be finalized by the CDR are still incomplete, such as the front wheel design, the finite element analysis on Abaqus for the fuselage ply book, and the potential weight reduction of the tail structure. It was recommended to complete these tasks as soon as possible by assigning different priorities to each, starting with the front wheel design, which, as an integral part of the overall design, must be addressed urgently.
- Assigned actions:

Action	Responsible	Deadline
Front-wheel design final definition	Lorenzo Martella	22nd December
Fuselage ply book finite element analysis	Giorgia Pallara	22nd Decemebr
Tail structure's weight optimization	Luigi Hayashida	17th December

• Topic 2: Requirements Review and Compliance Assessment

- Summary:

The team presented a Microsoft List outlining all the requirements, the corresponding regulation section, the specific paragraph, and a column for each department to report the measures taken to ensure compliance. Additionally, there was a column for any notes to be recorded during the Critical Design Review. The table below summarizes the content of the Microsoft List, including the notes taken during the meeting.

Structural and Construction Requirement Review Table				
Requirement	Regulation Reference	Actions Taken	Remaining Measures	Notes
Maximum Take Off Mass (MTOM) under 10.0 kg	4.1.2 Air vehicle Mass and CG	The structure was optimized with lightweight materials and iterative design modifications to achieve a target weight of 7 kg, identified through the Size Matrix Plot (SMP) method. A 20% margin was added to component mass estimates for unforeseen weight increases, with the final weight validated through direct measurement.	The BMC department continues to seek the lightest materials that meet mechanical requirements.	
Storage container - external dimensions of 1500 mm x 600 mm x 600 mm	4.5.1 Storage Container	A CAD model has been created for box design and positioning of parts within box dimensions has been studied. Compartment design for easy unloading and protection of drone parts	Finalize the design. Select the most suitable manufacturing technology, including materials and assembly procedure	
No tools routine assembly/disassembly	4.5.2 Assembly	Connections between the wing, fuselage, and tail are designed for tool-free assembly. The wing-fuselage uses a tube and pin, with horizontal translation constrained by drawbolts (under development). The tail is bonded to the boom and slides through the fuselage, secured by bulkheads and a spar.	Finalize the development of the drawbolts.	
Payload of at least 1.25 kg of sand	4.4.1 Payload - Aid Package Specification	The structure has been designed to minimize structural weight and maximize the payload weight. The payload is currently estimated at 2.5 kg		
Secondary retention mechanisms for major subassembly component connections	4.5.3 Connections	Secondary retention mechanisms have been provided for the major subcomponents.	Study whether it is possible to add additional safety mechanisms.	
Retention straps (payload) must be releasable without tools	4.4.2 Aid Package Carriage and Delivery	The payload release system features a pin holder on the tail boom, with a pin activated by a servo motor. The payload is secured with cables, one end attached to the pin and the other to the tail boom. After release, the cables will be stored in the fuselage. No tools are needed.		
Tracker with clear view of the sky, 140-degree field of view	4.4.3 GPS Tracker	A fibreglass and rohacell cover has been designed at the rear of the fuselage for aerodynamic reasons. It can be used to support electronic elements that could have problems staying in contact with carbon fibre. Inner shelves in balsa wood can be used to ensure the right positioning.		
Vehicle designed to carry an RF locator	4.4.4 RF Locator	A fibreglass and rohacell cover was designed at the rear of the fuselage for aerodynamic reasons. It can be used to support electronic elements such as the RF locator	Study the placement of the RF locator once more precise size information is available.	

Structural and Construction Requirement Review Table				
Requirement	Regulation Reference	Actions Taken	Remaining Measures	Notes
Batteries positively secured	4.2.1 Batteries	Removable two-compartment shelves have been designed to accommodate all the electronics. Batteries will be secured to the upper shelf.	The prevention of battery movements will be validated through some non-destructive tests on the final plane, as well as during the testing campaign.	Create a brief test report to provide evidence of the results obtained if needed.
Wiring retained securely	C.4 Construction Quality	During the construction of the aircraft, the wires will be placed in an ordered way, such that they will not run near critical spots such as in the payload bay, near mechanisms or in the proximity of the flight controls.	Tests will be done to ensure that the cables stay in place during the mission.	Create a brief test report to provide evidence of the results obtained if needed.
Batteries located such that it has some protection in the event of a crash	4.2.1 Batteries	Through the designed shelves of the electronics bay, the battery is on the upper and therefore the safest component and the most protected from crashes. It is also the closest to the faring and therefore easy and quick to remove with only two hands.		

Electronics Propulsion and Control Requirement Review Table				
Requirement	Regulation Reference	Actions Taken	Remaining Measures	Notes
Automatically navigating via GPS coordinates	4.1.7 Navigation System	This requirement has been satisfied by Ardupilot on a testing drone.	Validation during the testing campaign of the aircraft.	Create a brief test report to provide evidence of the results obtained if needed.
Vehicle capable of storing the coordinates of the Geo-fence	4.1.7 Navigation System	This requirement has been satisfied by Ardupilot on a testing drone.	Validation during the testing campaign of the aircraft.	Create a brief test report to provide evidence of the results obtained if needed.
Autonomous command signal to activate the Flight Termination System if breaking Geo-fence boundary	4.1.7 Navigation System	The requirement has been satisfied by Ardupilot on a testing drone, using a FTS sequence which did not destroy the plane on activation.	Same test on the final aircraft during the testing campaign, in order to validate the system.	Create a brief test report to provide evidence of the results obtained if needed.
Fly for 15 minutes with 20 per cent battery capacity remaining.	A.3.2 Mission Task: Payload Delivery	In the design phase, compliance with this requirement has been tested through simulation.	Validation during the testing campaign of the aircraft.	Create a brief test report to provide evidence of the results obtained if needed.
Audible and visual alert	4.1.10 Land-away recovery	This requirement has been satisfied by Ardupilot on a testing drone.	Validation during the testing campaign of the final aircraft.	Create a brief test report to provide evidence of the results obtained if needed.
Only batteries for energy storage	4.2.1 Batteries	One LiPo 6S battery have been selected to be the only energy storage device on the aircraft.		
Lithium Polymer (LiPo) batteries	4.2.1 Batteries	One LiPo 6S battery has been selected to be the only energy storage device on the aircraft.		
Bright colours in batteries	4.2.1 Batteries	The batteries will have some colourful tape on each side in order to be easily visible also in scarce-visibility environment.		
Clearly labelled link to isolate power to the whole aircraft	4.2.3 Propulsion Energy Safety Link	The safety switch to the aircraft will be placed on the bottom of the aircraft, away from the propeller, and will be easily accessible and labelled as specified in the requirement.		
Flown manually or automatically from take-off to full stop and payload release	4.3.1 Manual/autonomous control	The flight will be carried on by Ardupilot in autonomous mode, from take-off to landing. The possibility to take the manual control when necessary is guaranteed.	This will be validated during several tests on the airfield.	Create a brief test report to provide evidence of the results obtained if needed.
Revert to manual control at any point	4.3.1 Manual/autonomous control	The manual control and the FTS will be set on some radio switches. The functionality of these switches will be tested before each flight.		
Controller in Mode 2 configuration,	4.3.1 Manual/autonomous control	Radio will be set and kept on mode 2. This will be verified before each flight.		
FTS: – Pitch: full nose up – Roll: full right – Yaw: full right – High lift devices (if fitted): max. deflection – Spoilers/speed brakes (if fitted): max. deflection	4.3.2 Flight Termination System	The FTS will be correctly programmed in Ardupilot to execute the required sequence, eventually through the use of Lua scripts. No parachute has been designed. The failsafe will be validated before each flight.	The functionality of FTS will be validated through tests on ground and in-flight (while flying, without actually activating the sequence).	Create a brief test report to provide evidence of the results obtained if needed.

Electronics Propulsion and Control Requirement Review Table				
Requirement	Regulation Reference	Actions Taken	Remaining Measures	Notes
FTS activation: – Breach of Geo-fence. – After 5 seconds of RC Signal loss – After no longer than 10 seconds from lost Datalink. – Manually (RC Controller)	4.3.2 Flight Termination System	The FTS will be triggered by geo-fence breach, by signal loss, by datalink loss. It will be also possible to be manually triggered from both the radio and the GCS.	All the events which must activate the FTS (geo-breach, signal loss, manual trigger, ...) will be validated during the tests.	Create a brief test report to provide evidence of the results obtained if needed.
GCS: – Current air vehicle position on a map – Local Airspace, including the Geo-fence Flying Zone; – Height AGL (QFE); – Indicated Airspeed (kts); – Information on air vehicle Health.	4.3.3 Ground Control Station	The GCS will have a 2.4GHz link separated from the radio link, to ensure safety and redundancy in case of radio link loss. The GCS will be connected to a PC, which will display all the telemetry and will have the capability to control the aircraft (in particular, to trigger the FTS). The PC will be eventually connected to a tablet, in order to have a more portable device near to the pilot. The radio itself will be capable of showing the telemetry.		
Suitable port for wired connection between air vehicle and GCS	4.3.3 Ground Control Station	The USB interface + buzzer will be present on the flight controller and easily accessible on the plane. To connect to the PC, a USB cable will always be present in the box.		
Radio equipment, ranges of at least 1 km	C.5 Radio Equipment	The chosen 2.4GHz true-diversity RC receiver and the radio will run ExpressLRS, a protocol that demonstrated to ensure one of the best performances on the market in terms of signal quality and range. Two antennas will be placed on the bottom of the plane, outside the fuselage, one for each side, to ensure full coverage during the mission.	The range of the radio link will be validated during the tests, bringing the radio at 1km or more from the plane and verifying the power of the signal.	Create a brief test report to provide evidence of the results obtained if needed.
FTS and radio equipment 'Spread Spectrum' compliant on the 2.4 GHz band	4.3.4 Radio Equipment	2.4GHz true-diversity RC receiver has been chosen.		
Electric motor	C.1 Concept Selection	The selected propulsion system of the drone includes a T-Motor AT4120 KV560, which satisfies the requirement.		
Geo-Fence with margin to allow for drift	C.8 Route planning around Way-points	Before each flight, a simulation of the flight path will be carried on Mission Planner, applying some corrections where necessary. The path shall never go too close to the geo-fence. A margin due to errors or weather will be considered on top of the simulation.		
Aid Package deployed by automatic command	4.4.2 Aid Package Carriage and Delivery	Payload Release system is designed to be activated by an automatic command.	Validation during the testing campaign of the aircraft, to verify compliance of payload parachute and declared values.	Create a brief test report to provide evidence of the results obtained if needed.

Financial Requirement Review Table				
COTS (Commercial Off the Shelf) costs under £1300	4.7 Limits on the use of COTS Items	During the design of the aircraft, a spreadsheet with the cost of each COTS has been kept updated to ensure compliance with the requirement.	BMC department is still looking for sponsorships for carbon tubes and cameras.	
Secondary Requirement Review Table				
Fuselage marked empty CG	4.1.2 Air vehicle Mass and CG	Precise CG position has been calculated and the BMC department will ensure to have a specific sticker.	Once the aircraft is built, the actual CG position will be manually verified.	
Bright colours	4.1.10 Land-away recovery	The BMC department is designing a livery to meet this requirement.		

Operational Requirement Review Table				
Requirement	Regulation Reference	Actions Taken	Remaining Measures	Notes
Take-off in 10 m x 10 m box	4.1.6 Take-off and Landing	Take-off requirement has been considered in the Size Matrix Plot. The design point found satisfies the demand.		
Landing in 10 m x 20 m box	4.1.6 Take-off and Landing	Flaps have been designed with the AER department in compliance with the landing requirements.		
Max 200 ft AGL	4.6.5 Operational Safety Requirements	A geo-fence on altitude will be imposed on Ardupilot, and the telemetry will always be kept under the sight of the pilot.		
Max 40 kts true airspeed (KTAS)	4.1.4 Air vehicle fly-off envelope limitations	The maximum airspeed will be imposed on Ardupilot. The functionality of the speed limit of Ardupilot has been proved on a testing drone.	Validation during the testing campaign of the aircraft.	Create a brief test report to provide evidence of the results obtained if needed.
Max bank angle around 30 degrees	C.8 Route planning around Way-points	The maximum bank angle will be imposed in Ardupilot, as well as the navigation parameters will be tuned accordingly to the capabilities of the aircraft.		
Operate in winds of up to 20 kts gusting to 25 kts, and light rain	4.1.5 Weather Limitations	A V-tail with a low butterfly angle (38°) will guarantee low wind sensibility and great static and dynamic stability.		
Take-off and landing in crosswind of 5 kts with gusts of 8 kts	4.1.5 Weather Limitations	To minimize the aircraft's susceptibility to gusts of wind, vertical projection of the V-tail has been minimized while maintaining good stability parameters.		
Vehicle capable of being flown again after landing with fresh battery but without repairs	4.1.6 Take-off and Landing	During tail boom sizing the possibility of a tail strike during take-off and landing has been considered to ensure that the tail will not hit the ground during take-off rotations and landing flares.		
Vehicle designed to operate on hard or grass runways	4.1.6 Take-off and Landing	The type of runway has been considered in the design of the landing gear, thanks to the experience gained from past models.		
Aid Package deployed during flight	4.4.2 Aid Package Carriage and Delivery	The payload release system has been designed to be activated only during flight.		
Assemble and pre-flight the UAS out of the Storage Container within 6 minutes	5.3.5 Storage and Flight Readiness Demonstration	The aircraft's structure, particularly the joints between the different parts, has been designed to be as quick and intuitive as possible.	The assembly and flight preparation time will be tested multiple times once the aircraft is built.	
Catapults set up during the Pre-Flight Preparation	5.3.2 Set up of Catapults	No catapults will be needed		
Take-off preparation within 5 minutes	A.2.4 Final Check-out	Alongside an easy-to-assemble structure, checklists for pre-flight checks will be created.		
Max 15 minutes of flight	A.3.2 Mission Task: Payload Delivery	In the design phase, the compliance with this requirement has been tested through simulation.	Validation during the testing campaign of the aircraft.	Create a brief test report to provide evidence of the results obtained if needed.
Flight within Visual Line of Sight (VLOS) from the Pilot	4.6.5 Operational Safety Requirements	The mission will be scheduled to take place within the pilot's line of sight. In case of aircraft loss, the FTS will always be activated in accordance with the corresponding requirements.		

- **Topic 3: Reviewer's Feedback and Recommendations**

- Summary: The analysis of each individual requirement has been thorough, and the executed components have been well managed. It is recommended to track all tests conducted as references for future years. For instance, in the radio range test, a brief report should be prepared, including a map with the location of the radio, aircraft, and the results obtained. This will provide evidence in case any competitor requests proof of the test. Tracking for validation is essential.

Regarding the management of requirements, no further points need to be added; they have been effectively handled.

The presentation was effective, given the project's complexity, the multiple components involved, and the tight timelines. The information was concise and clear. The next step is to manage the open issues, assign deadlines and priorities to each, and monitor their progress until resolution. These issues and their respective updates should be documented in the meeting minutes for accountability. Tracking the progress of open issues post-presentation is crucial.

It is suggested to create a shared table with assigned responsibilities and deadlines for each action item.

Next Steps

- **Tasks:**

The team is committed to addressing the reviewer's recommendations as outlined in the "Assigned Action" section of "Topic 1" and in the "Note" column of the Requirement Review Tables. The team also commits to providing updates on project developments and to promptly communicate any technical or organizational issues to facilitate a timely discussion of possible resolutions.

Closure

Prepared by:



(Team Leader)

Approved by:

(Reviewer)

On: 13/12/2024